

Laboratory 1

GUI (Graphical User Interface) in Java

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Activity 1 – Question 2

Write a Java program which creates two frames, each containing one button. When you execute the program only one JFrame will appear, and when you click on its button it will disappear and the other JFrame will appear. If you click on the button of the second frame, it will make it disappear and let the first frame appear. And so on.

Java Code

MyFrame1.java

```
import javax.swing.JButton;
import javax.swing.JFrame;
import java.awt.GridBagLayout;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

// MyFrame1 - first frame with a "click" button.
// Clicking the button hides this frame and shows MyFrame2.
public class MyFrame1 extends JFrame implements ActionListener {

    private JButton clickButton;
    private MyFrame2 frame2;

    public MyFrame1() {
        setTitle("MyFrame 1");
        setSize(320, 240);
        setLocationRelativeTo(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // GridBagLayout with no constraints keeps the button small
        // and centers it in the middle of the frame.
        setLayout(new GridBagLayout());

        clickButton = new JButton("click");
        clickButton.addActionListener(this);
        add(clickButton);
    }

    // Give this frame a reference to the other frame so it can switch to it.
    public void setFrame2(MyFrame2 frame2) {
        this.frame2 = frame2;
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        this.setVisible(false);
        frame2.setVisible(true);
    }
}
```

MyFrame2.java

```
import javax.swing.JButton;
import javax.swing.JFrame;
import java.awt.GridBagLayout;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

// MyFrame2 - second frame with a "click" button.
// Clicking the button hides this frame and shows MyFrame1.
public class MyFrame2 extends JFrame implements ActionListener {

    private JButton clickButton;
    private MyFrame1 frame1;

    public MyFrame2() {
        setTitle("MyFrame 2");
    }
}
```

```

        setSize(320, 240);
        setLocationRelativeTo(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        setLayout(new GridBagLayout());

        clickButton = new JButton("click");
        clickButton.addActionListener(this);
        add(clickButton);
    }

    // Give this frame a reference to the other frame so it can switch to it.
    public void setFrame1(MyFrame1 frame1) {
        this.frame1 = frame1;
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        this.setVisible(false);
        frame1.setVisible(true);
    }
}

```

Activity1Main.java

```

// Activity 1 - Question 2
// Creates two JFrames, each with one "click" button.
// Only MyFrame 1 is visible at start. Clicking its button swaps
// visibility to MyFrame 2, and clicking MyFrame 2's button swaps back.
public class Activity1Main {
    public static void main(String[] args) {
        MyFrame1 frame1 = new MyFrame1();
        MyFrame2 frame2 = new MyFrame2();

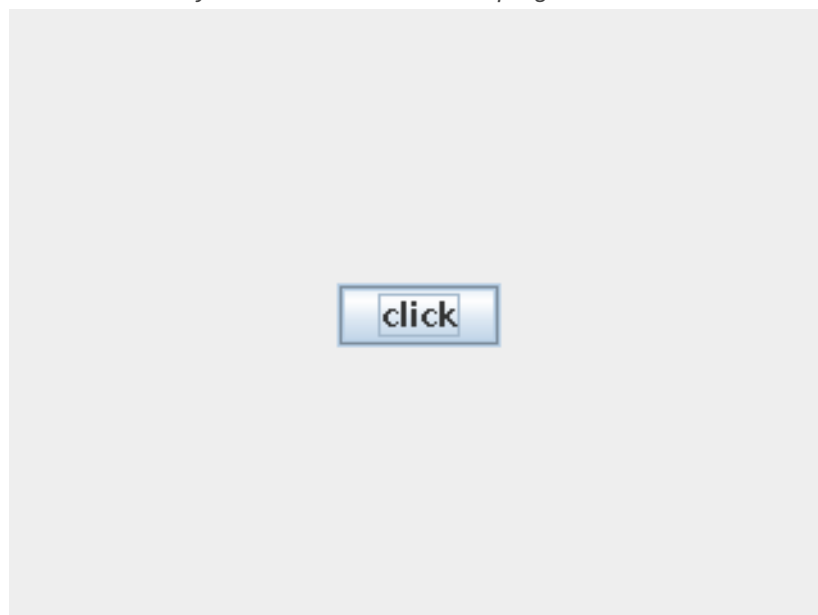
        frame1.setFrame2(frame2);
        frame2.setFrame1(frame1);

        // Only MyFrame 1 is visible when the program starts.
        frame1.setVisible(true);
    }
}

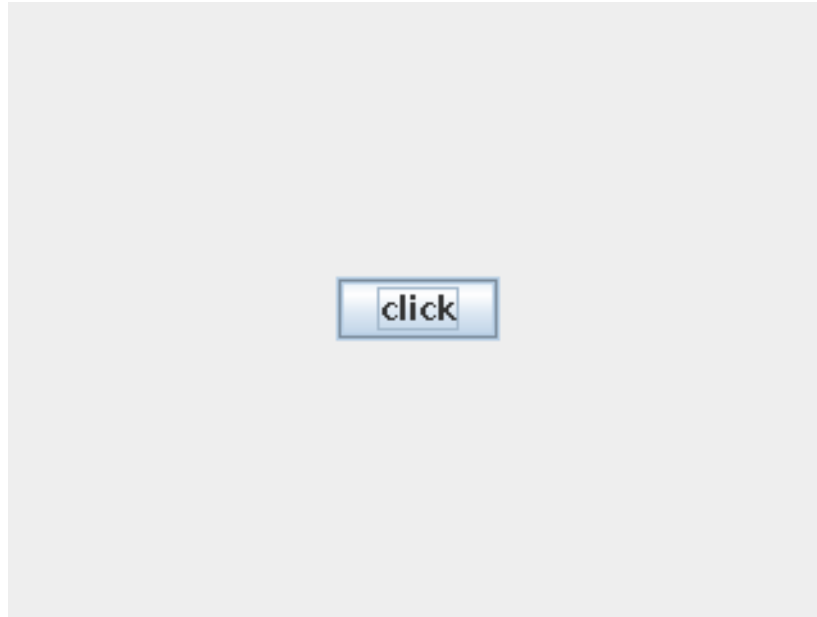
```

Output Screenshot

MyFrame 1 visible when the program starts



MyFrame 2 visible after clicking the button on MyFrame 1



Activity 2 – Question 1

Using the FlowLayout, write the program that creates the Distance Converter interface. On the click of button **Convert!**, the integer value in the text field should be converted to kilometers and displayed in the text area below, in a new line. (1 mile = 1.60934 kilometers)

Java Code

DistanceConverterFlow.java

```
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JScrollPane;
import javax.swing.JTextArea;
import javax.swing.JTextField;
import java.awt.FlowLayout;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.math.BigDecimal;

// Activity 2 - Question 1
// Distance Converter built with FlowLayout only.
// Converts miles entered by the user into kilometers.
public class DistanceConverterFlow extends JFrame implements ActionListener {

    private JLabel milesLabel;
    private JTextField milesField;
    private JButton convertButton;
    private JTextArea resultArea;

    // 1 mile = 1.60934 kilometers
    private static final BigDecimal MILES_TO_KM = new BigDecimal("1.60934");

    public DistanceConverterFlow() {
        setTitle("Distance Converter");
        setSize(400, 300);
        setLocationRelativeTo(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // The whole frame uses FlowLayout.
        setLayout(new FlowLayout());

        milesLabel = new JLabel("Distance in miles:");
        milesField = new JTextField(8);
        convertButton = new JButton("Convert!");
        convertButton.addActionListener(this);

        resultArea = new JTextArea(8, 24);
        resultArea.setEditable(false);
        JScrollPane scrollPane = new JScrollPane(resultArea);

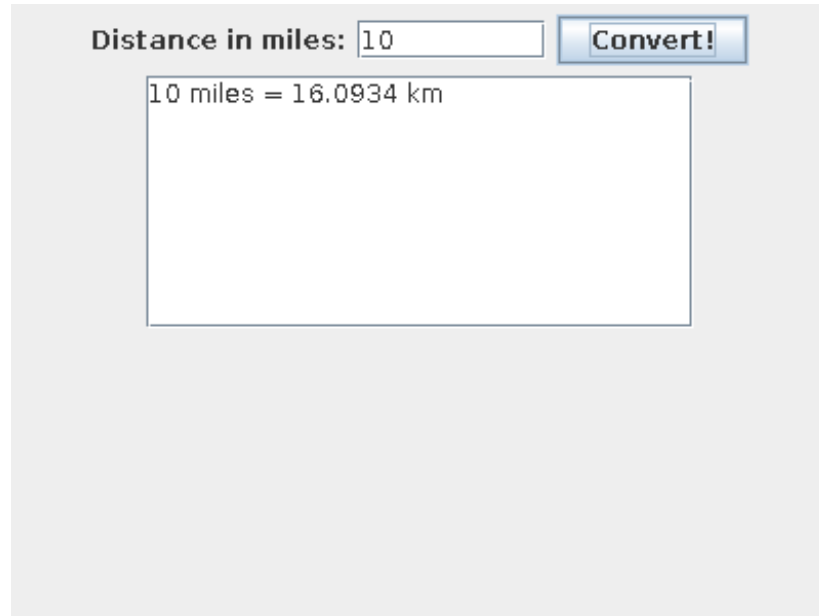
        add(milesLabel);
        add(milesField);
        add(convertButton);
        add(scrollPane);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        try {
            int miles = Integer.parseInt(milesField.getText().trim());
            BigDecimal km = new BigDecimal(miles).multiply(MILES_TO_KM);
            String kmText = km.stripTrailingZeros().toPlainString();
            resultArea.append(miles + " miles = " + kmText + " km\n");
        } catch (NumberFormatException ex) {
            resultArea.append("Please enter a valid integer.\n");
        }
    }
}
```

```
}  
  
public static void main(String[] args) {  
    DistanceConverterFlow frame = new DistanceConverterFlow();  
    frame.setVisible(true);  
}  
}
```

Output Screenshot

Input: 10 | Output: 10 miles = 16.0934 km



The screenshot shows a Java Swing window titled "Distance Converter". Inside the window, there is a label "Distance in miles:" followed by a text input field containing the number "10". To the right of the input field is a button labeled "Convert!". Below these elements is a large text area that displays the result: "10 miles = 16.0934 km".

Activity 2 – Question 2

Using a combination of two layout managers (BorderLayout and FlowLayout), modify the previous program to obtain the same interface. Place the label, text field and button in a FlowLayout panel positioned in the NORTH of a BorderLayout, and place the text area in the CENTER. On the click of button **Convert!**, the integer value in the text field should be converted to kilometers and displayed in the text area below, in a new line.

Java Code

DistanceConverterBorderFlow.java

```
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JPanel;
import javax.swing.JScrollPane;
import javax.swing.JTextArea;
import javax.swing.JTextField;
import java.awt.BorderLayout;
import java.awt.FlowLayout;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.math.BigDecimal;

// Activity 2 - Question 2
// Distance Converter built with a combination of BorderLayout and FlowLayout.
// NORTH: a FlowLayout panel with the label, text field and button.
// CENTER: the text area showing the conversion results.
public class DistanceConverterBorderFlow extends JFrame implements ActionListener {

    private JLabel milesLabel;
    private JTextField milesField;
    private JButton convertButton;
    private JTextArea resultArea;

    // 1 mile = 1.60934 kilometers
    private static final BigDecimal MILES_TO_KM = new BigDecimal("1.60934");

    public DistanceConverterBorderFlow() {
        setTitle("Distance Converter");
        setSize(400, 300);
        setLocationRelativeTo(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // The main frame uses BorderLayout.
        setLayout(new BorderLayout());

        // NORTH panel uses FlowLayout for the label, text field and button.
        JPanel northPanel = new JPanel(new FlowLayout());
        milesLabel = new JLabel("Distance in miles:");
        milesField = new JTextField(8);
        convertButton = new JButton("Convert!");
        convertButton.addActionListener(this);

        northPanel.add(milesLabel);
        northPanel.add(milesField);
        northPanel.add(convertButton);

        // CENTER area shows the conversion results.
        resultArea = new JTextArea();
        resultArea.setEditable(false);
        JScrollPane scrollPane = new JScrollPane(resultArea);

        add(northPanel, BorderLayout.NORTH);
        add(scrollPane, BorderLayout.CENTER);
    }
}
```

```

@Override
public void actionPerformed(ActionEvent e) {
    try {
        int miles = Integer.parseInt(milesField.getText().trim());
        BigDecimal km = new BigDecimal(miles).multiply(MILES_TO_KM);
        String kmText = km.stripTrailingZeros().toPlainString();
        resultArea.append(miles + " miles = " + kmText + " km\n");
    } catch (NumberFormatException ex) {
        resultArea.append("Please enter a valid integer.\n");
    }
}

public static void main(String[] args) {
    DistanceConverterBorderFlow frame = new DistanceConverterBorderFlow();
    frame.setVisible(true);
}
}

```

Output Screenshot

Input: 10 | Output: 10 miles = 16.0934 km

Distance in miles:

10 miles = 16.0934 km